

Mathews's Algorithm

From the reader's point of view, the existence in literature of potentiality in its Oulipian sense has the charm of introducing duplicity into all written texts, whether Oulipian or not. It isn't merely a sonnet in Queneau's *100,000 Billion Poems* on which doubt is cast by the horde of alternatives waiting to take its place: the most practical work of prose, no matter how sturdy it may seem in its apparent uniqueness, will prove just as fragile as soon as one thinks of subjecting it to the procedures of $S + 7$ or Semo-Definitional Literature.¹ Beyond the words being read, others lie in wait to subvert and perhaps surpass them. Nothing any longer can be taken for granted; every word has become a banana peel. The fine surface unity that a piece of writing proposes is belied and beleaguered; behind it, in the realm of potentiality, a dialectic has emerged.

The algorithm here discussed is a new means of tracking down this otherness hidden in language (and, perhaps, in what language talks about). It is one more way of saying two things at once. It has its particular attractions. First of all, it's a simple mechanism into which complex materials can be introduced. If these materials require a certain amount of care in their presentation to form an algorithmic table, their potential duplicity is realized through means that are virtually automatic. The algorithm can make use of existing material as well as of material specially invented for it (a point that will be abundantly demonstrated later on). Its creative potentiality can manifest itself either in the actual construction of the table or in its subsequent solution. It can be used both to decompose (or analyze) texts or to compose (or invent) them. Last, the algorithm can be applied to every one of the rows of Queneau's Table, "semantic" as well as "syntactic." (And why stop there? The limitations adopted here—which run from letters to rudimentary semantic elements—define only a small part of what the algorithm can handle. It is capable of dealing with *fragments* of letters, either graphic or phonetic, as well as *their* component

parts, not to mention amoebas, molecules, and quarks. It can juggle not only episodes of fiction, as will be shown later, but entire books, indeed entire literatures and civilizations, planets, solar systems, galaxies—anything in fact that can be manipulated either in its material or its symbolic form. . . .)

So: how does it work?

Here is what you do.

I. Materials: Several—at Least Two—Sets of Heterogeneous Elements Are Required

Comment: The heterogeneity can sometimes be minimal, but it is nonetheless essential for the proper functioning of the algorithm. The less it is respected, the more likely the machine will simply reproduce replicas of the original sets. In theory, to avoid any chance of this happening, the rule is that in a table of n elements, $n^2 - (n - 2)$ elements must be different. (Example: in a table of 4 sets of 4 elements, 14 of the 16 elements must differ.) In practice, however, it is often possible to lower this level of heterogeneity.

II. Arrangement

1. All sets must contain the same number of elements.
2. Each element in a set must have a function equivalent to or consistent with the corresponding elements of the other sets. (If, for instance, the elements are words, the equivalence will be one of grammatical function.) In other words, the elements of each set must be arranged in the same order. (Not $a_1b_1c_1d_1$ and $c_2a_2d_2b_2$ but $a_1b_1c_1d_1$ and $a_2b_2c_2d_2$.)
3. The sets are superimposed one above the other to form a table consisting of rows (the sets) and columns (the corresponding elements). Here is a table of 4 sets of 4 elements:

1	a_1	b_1	c_1	d_1
2	a_2	b_2	c_2	d_2
3	a_3	b_3	c_3	d_3
4	a_4	b_4	c_4	d_4

Comment: In all the examples, the number of elements in each set equals the number of sets. This is a convenience, not a requirement.

III. Operation

1. Shift each set $n - 1$ places left. (In other words, shift the second set one place left, the third two places, etc.)

1	a_1	b_1	c_1	d_1
2	b_2	c_2	d_2	a_2
3	c_3	d_3	a_3	b_3
4	d_4	a_4	b_4	c_4

2. Read the columns downward starting with the initial elements (a) of the sets. Four new sets result:

a_1	b_2	c_3	d_4
a_4	b_1	c_2	d_3
a_3	b_4	c_1	d_2
a_2	b_3	c_4	d_1

3. Perform the same manipulation with a shift *right*:

1	a_1	b_1	c_1	d_1
2	d_2	a_2	b_2	c_2
3	c_3	d_3	a_3	b_3
4	b_4	c_4	d_4	a_4

4. Read the results upward, still beginning with the initial elements (a) of the original sets:

a_1	b_4	c_3	d_2
a_2	b_1	c_4	d_3
a_3	b_2	c_1	d_4
a_4	b_3	c_2	d_1

In this way, from n sets $2n$ sets can be obtained.

IV. Uses

The applications of the algorithm considered here are classified according to the *elements* forming the sets: a , b , c , etc. will be in turn letters, words, word groups, groups of word groups, and, finally, semantic elements. At every stage, two examples will be given, one involving existing material, the other new material.

1. Letters

(a) Using existing material: here, the words of all known languages, living or dead. From English:

T	I	N	E
S	A	L	E
M	A	L	E
V	I	N	E

After a shift left:

T	I	N	E
A	L	E	S
L	E	M	A
E	V	I	N

(tale, vile, mine, sane)

After a shift right:

T	I	N	E
E	S	A	L
L	E	M	A
I	N	E	V

(tile, sine, mane, vale)

(b) Using new material, i.e., words that do not yet exist, in known or imaginary languages.

This example is drawn from the dialect of a Montagnard tribe in Pan-Nam (cf. H. Mathews, *The Sinking of the Odradek Stadium*) that uses a vocabulary at once spontaneous and combinatorial:

B	A	G	O
K	E	D	U
F	E	S	I
L	O	M	I

After a shift left:

B	A	G	O
E	D	U	K
S	I	F	E
I	L	O	M

(besi, ladi, fogu, kemo)

After a shift right:

B	A	G	O
U	K	E	D
S	I	F	E
O	M	I	L

(bosu, kami, fegi, ledi)

2. Words

Here the rule of equivalence (II, 2) becomes obligatory. Sets of words² must be arranged in identical grammatical sequence.

(a) Using existing material: take the first lines of four well-known sonnets of Shakespeare:

Shall I compare thee to a summer's day?
 Music to hear, why hear'st thou music sadly?
 That time of year thou may'st in me behold . . .
 Farewell! thou art too dear for my possessing . . .

Arrangement: Reduce each line to four main words (the remaining words will be considered as their auxiliaries):

I, compare, summer, day (shall, thee, to a)
 music, hear, thou, music (why, sadly)
 time, year, thou, behold (that, of, in me)
 farewell, thou, art(dear), possessing (too, for my)

Align these four sets according to corresponding parts of speech:

I summer day compare (shall, thee, to a)
 thou music music hear (why, sadly)
 thou time year behold (that, of, in me)
 thou farewell possessing art (dear) (too, for my)

Apply a shift left to this table. The results will read as:

I music year art (dear)
 thou summer music behold
 thou farewell day hear
 thou time possessing compare

Reintroducing the auxiliaries, compose four new lines in blank verse (the openings of four undiscovered Shakespearean sonnets):

The music of the years, too dear for me . . .
 The summer's music sadly thou beholdest . . .
 Today thou may'st in me hear the farewell . . .
 Possessing thee, why doth my Time compare . . .

Similarly, after a shift right:

Shall I, sadly hearing the year's farewell . . .
 Possessing me, thou hast beheld my summer . . .
 Too dear a music for a day thou art . . .
 Music thou art of Time, that doth compare thee . . .

(b) Using new material: in the following example, where words are taken one by one, four propositions are arranged according to the rules of the algorithm:

Truth left him cold.
 Wealth made her glad.
 Work turned you sour.
 Love kept me free.

After a shift left:

Truth made you free.
 Love left her sour.
 Work kept him glad.
 Wealth turned me cold.

After a shift right:

Truth kept you glad.
 Wealth left me sour.
 Work made him free.
 Love turned her cold.

Extending the notion of word to include auxiliaries (cf. IV, 2, a), we can move on to a more elaborate table:

Frustrated members	of the Cabinet	were found conspiring in	local massage parlors.
Drivers	from the Sanitation Department	will go on strike against	the new garbage helicopters.
Six terrorists	of the Keep Kids Kleen Klub	have planted bombs in	sexually integrated rest-rooms.
Female impersonators	not licensed by their union	are not allowed into	nursing schools.

After a shift left we read:

Frustrated members	from the Sanitation Department	have planted bombs in	nursing schools.
Female impersonators	of the Cabinet	will go on strike against	sexually integrated rest-rooms.

Six terrorists	not licensed by their union	were found conspiring in	the new garbage helicopters.
Drivers	of the Keep Kids Kleen Klub	are not allowed into	local massage parlors.

And after a shift to the right:

Frustrated members	not licensed by their union	have planted bombs in	the new garbage helicopters.
Drivers	of the Cabinet	are not allowed into	sexually integrated rest-rooms.
Six terrorists	from the Sanitation Department	were found conspiring in	nursing schools.
Female impersonators	of the Keep Kids Kleen Klub	will go on strike against	local massage parlors.

3. Word groups (verses, sentences)

(a) Using existent material: returning to Shakespeare, take 14 of his best known sonnets, arrange the verses of each sonnet horizontally so as to form a set, and superimpose the sets. After a shift left, here is the first of the 14 possible readings (slightly edited as to punctuation):

Shall I compare thee to a summer's day
 And dig deep trenches in thy beauty's field?
 Why lov'st thou that which thou receiv'st not gladly,
 Bare ruin'd choirs where late the sweet birds sang?
 Anon permit the basest clouds to ride
 And do what'er thou wilt, swift-footed Time:
 Nor Mars his sword, nor war's quick fire, shall burn
 Even such a beauty as you master now.
 Love's not Time's fool, though rosy lips and cheeks
 (When other petty griefs have done their spite,
 And heavily) from woe to woe tell oe'r
 That Time will come and take my love away;
 For thy sweet love remembered such wealth brings
 As any she belied beyond compare.

Comment: It is perhaps at this point that the analytic potential of the algorithm clearly manifests itself. This new poem throws light on the structure and movement of the Shakespearean sonnet.

(b) Using new material: the following example demonstrates a table made up of 3 sets of 3 sentences. Each sentence refers to an event concerning three different characters.

The conductor wanted his orchestra to produce an utterly muffled sound.	He declared, "If we keep everything low-key, perhaps we can work out a satisfactory arrangement."	He kept insisting: "Please don't strike the drums with anything harder than a sponge!"
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The young Laotian chiropodist wandered around New York with his phrase book, which was a fraud—a collection of useless, ridiculous expressions.	He kept repeating his request without ever managing to get what he wanted: an adequate supply of absorbent cotton.	It wasn't until he flashed a fat bank-roll that a compassionate saleswoman at last took care of him.
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A crafty sheik came to town to buy a wife on favorable terms.	He tried Macy's, Bloomingdale's, and Korvette's without finding what he needed.	Since it was hard to take him seriously, everyone simply ignored his words.
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After a shift left:

The conductor wanted his orchestra to produce an utterly muffled sound.	He kept repeating his request without ever managing to get what he wanted: an adequate supply of absorbent cotton.	Since it was hard to take him seriously, everyone simply ignored his words.
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4. Groups of word groups (stanzas, paragraphs, episodes, etc.)

(a) Using existing material: Shakespeare is here joined by George Herbert, Ben Jonson, and John Donne.

Take four sonnets that follow the Shakespearean model and divide them into their four main components: three quatrains and the concluding couplet. Using these components as elements, align those of each poem horizontally to form a set. Arrange the sets as a table.

Here, after a shift left, is the first of four possible readings:

Farewell! thou art too dear for my possessing,
And like enough thou know'st thy estimate:
The charter of thy worth gives thee releasing,
My bonds in thee are all determinate,
While mortal love doth all the title gain!
Which siding with invention, they together
Bear all the sway, possessing heart and brain
(Thy workmanship) and give thee share in neither.
But now thy work is done, if they that view
The several figures languish in suspense
To judge which passion's false, and which is true,
Between the doubtful sway of reason and sense;
That thou remember them, some claim as debt;
I think it mercy if thou wilt forget.

(b) Using new material: here a process of condensation can be profitably introduced. Rather than manipulate lengthy paragraphs, it seems more interesting to apply the algorithm directly to more substantial entities. The elements in the following examples are fictional episodes, summed up in one or two sentences.

1. (a) A and B are husband and wife. (b) A is summoned to the bedside of his ailing mother. (c) At this very moment B becomes involved with someone else. (d) On his return, A feels disgust for B, whom he decides to exclude from his life. (e) B has a nervous breakdown.

2. (a) A loves B. B does not love A. (b) When he inherits a fortune, A at first thinks of letting B benefit from it. (c) In the meantime, B, who has fallen on evil days, starts thinking of A as a possible savior. (d) If B now looks on A with fresh eyes, A, absorbed in his new life, is gradually forgetting about B altogether. (e) A and B never see each other again.
3. (a) A owes B a great deal of money. (b) Having decided he must earn financial independence, A settles in a developing country, where he succeeds in amassing a fortune. (c) B's mother gives B an unexpected sum of money. She also tells B of the goal A has set himself. (d) These new circumstances move B to suggest to A that he forget his former obligations. (e) A never forgives B for nullifying the solicitude of so many years.
4. (a) A knows young B well, but he does not know that he is B's father. B is also ignorant of the fact. (b) A accidentally learns that he is the father of a child whom he has never known. He drops everything to look for his offspring. (c) B's mother falls ill; B is notified. (d) B thus finds A at the ailing woman's bedside. B's unexpected appearance is the immediate cause of her death. (e) The relationship between A and B remains unchanged.
5. (a) A and B know each other. A decides to kill B in order to experience an Existential "random act." (b) A falls in love with an acquaintance of B. (c) Quite involuntarily, B becomes violently resentful of this person for taking A away. (d) Furious at B's outrageous behavior, A abandons his plan. But A's anger starts B wondering if he isn't considering murder. (e) B kills A.

After a shift right:

1. (a) A and B are husband and wife. (b) A falls in love with an acquaintance of B. (c) B's mother falls ill; B is notified. (d) These new circumstances move B to suggest to A that he forget his former obligations. (e) A and B never see each other again.
2. (a) A loves B. B does not love A. (b) A is summoned to the bedside of his ailing mother. (c) Quite involuntarily, B becomes furiously resentful of this person for taking A away. (d) B thus finds A at the ailing woman's bedside. B's unexpected appearance is the immediate cause of her death. (e) A never forgives B for nullifying the solicitude of so many years.
3. (a) A owes B a great deal of money. (b) When he inherits a fortune, A at first thinks of letting B benefit from it. (c) At this very

moment B becomes involved with someone else. (d) Furious at B's outrageous behavior, A abandons his plan. But A's anger starts B wondering if he isn't considering murder. (e) The relationship between A and B remains unchanged.

4. (a) A knows young B well, but he does not know that he is B's father. B is also ignorant of the fact. (b) Having decided he must earn financial independence, A settles in a developing country, where he succeeds in amassing a fortune. (c) In the meantime, B, who has fallen on evil days, starts thinking of A as a possible savior. (d) On his return, A feels disgust for B, whom he decides to exclude from his life. (e) B kills A.

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5. Semantic elements

The OuLiPo divides constrictive structures into *syntactic* and *semantic*. The former affect the material aspects of language (letters, words, syntax); the latter affect what language talks about (subject, content, meaning). Obviously, in the last example we have already moved from syntactic to semantic objects. Perhaps it will suffice to speed the algorithm on its course toward the absolute by ending this demonstration with one or two examples in which the semantic factor is even stronger.

(a) Using existing material: in this case, *Hamlet*. We can sum up our perception of the play according to the rules of the algorithm:

- | | | | | |
|-----------------------------------|--------------------|-------------|--------------------------------|------------------------------------|
| 1. Having received instructions, | the son | hesitantly | chooses revenge. | Death is the condition of success. |
| 2. Irrepressibly erotic, | the widowed mother | hastily | remarries. | This is a deceptive pleasure. |
| 3. In the clutches of resentment, | the uncle | desperately | indulges a posthumous passion. | Can we possibly approve? |

- | | | | | |
|--|--------------|-------------|-----------------------------------|------------------------------------|
| 4. Royally willful, | the dead man | obsessively | tries to re-write history. | Someone has to pay. |
| 5. A prey to this incestuous violence, | Ophelia | faithfully | abandons her reason and her life. | A father's death is also involved. |

After a shift to the right:

- | | | | | |
|--|--------------------|-------------|-----------------------------------|------------------------------------|
| 1. Having received instructions, | Ophelia | obsessively | indulges a posthumous passion. | This is a deceptive pleasure. |
| 2. Irrepressibly erotic, | the son | faithfully | tries to re-write history. | Can we possibly approve? |
| 3. In the clutches of resentment, | the widowed mother | hesitantly | abandons her reason and her life. | Someone has to pay. |
| 4. Royally willful, | the uncle | hastily | chooses revenge. | A father's death is also involved. |
| 5. A prey to this incestuous violence, | the dead man | desperately | remarries. | Death is the condition of success. |

Here is another way of handling the same subject:

- | | | | |
|---------------------------|------------|-----------------|---------------------|
| 1. With the husband dead, | the mother | marries | the brother-in-law. |
| 2. With the father dead, | the son | chooses | the mother. |
| 3. With Ophelia dead, | reason | takes | a holiday. |
| 4. With the brother dead, | the uncle | falls victim to | the son. |

After a shift right:

- | | | | |
|---------------------------|------------|-----------------|-------------|
| 1. With the husband dead, | the uncle | takes | the mother. |
| 2. With the father dead, | the mother | falls victim to | a holiday. |

- | | | | |
|---------------------------|---------|---------|---------------------|
| 3. With Ophelia dead, | the son | marries | the son. |
| 4. With the brother dead, | reason | chooses | the brother-in-law. |

Comment: It is probably unnecessary to point out that, here as elsewhere, the results yielded by existing works can be used in two ways: either as a means of commenting on those works or as materials for inventing new ones.

(b) Using new material: for this last example, the elements are far more abstract (although the abstraction falls short of actual concept); they represent situations or events that writers have used since the beginning of literature. They have been arranged in patterns of minimal causality.

Given love,	consummation;	therefore resumption.
Given possession,	danger;	therefore flight.
Given victory,	war;	therefore reassessment.

After a shift left:

Given love,	danger	therefore reassessment.
Given victory,	consummation;	therefore flight.
Given possession,	war;	therefore resumption.

After a shift right:

Given love,	war;	therefore flight.
Given possession,	consummation;	therefore reassessment.
Given victory,	danger;	therefore resumption.

Comment: Perhaps more clearly than the preceding ones, this example shows how the inventiveness of the algorithm (indeed, of Oulipian techniques in general) can be realized in two ways—(1) in the setting up of the constrictive structure and (2) in its solution. In most of the earlier examples, the algorithm has produced its effect by requiring the creation of a table that “works” (cf. what is involved in writing a snowball). In the last example, on the other hand, one’s ingenuity must be exerted to complete or justify the table (cf. what is involved in working out a procedure like “The Relation X takes Y for Z”). This last table is a table of *provocation*.

It must be realized that whatever its semantic level, the material of the algorithm must always be manipulated syntactically. (In the examples drawn from *Hamlet*, the elements are *handled* as words or phrases, no

matter what their meaning is.) The algorithm can approach abstraction only through simple linguistic systems.

And also, perhaps, through its actual operation: it is here, at any rate, that the “meaning” of the algorithm is to be found. This meaning is nothing new as far as the activity of the OuLiPo is concerned. First of all, two other Oulipian structures can be classed as special cases of the algorithm (that is, with sets composed of two elements): the perverbial poem and the equivoque (or “Poem for Moebius Strip”). Furthermore, the algorithm fits in with the OuLiPo’s general interest in combinatorial literature: the results the algorithm produces are, after all, only segments of the much vaster range of circular permutations that each of its tables implies. The algorithm would thus have as its remote ancestor the *Ars Magna* of Ramon Lull (which inspired Giordano Bruno’s admiration), Leibnitz’s *Dissertatio de Arte Combinatoria*, and Martin Gardner’s genuine, albeit skeptical, interest. In Lull’s work, circular permutations are given literal shape in figures consisting of concentric circles. These circles are divided into sections; by turning, each circle is made to form with its neighbors all the combinations possible between their respective sections. If most of Lull’s figures contain only two circles, one of them—the *figura universalis*—has as many as fourteen.

Fourteen concentric circles: wouldn’t Lull then become a forerunner of one of the most extraordinary works the OuLiPo has produced, the *100,000 Billion Poems* of Raymond Queneau? It is perfectly possible to inscribe Queneau’s sonnets on the *figura universalis*. Each circle need simply be divided into ten sections. The outermost circle would contain the first verses of the poems, the circle next to it the second verses, and so on. The arrangement would provide as complete a means of reading the work as it finds in book form (and, probably, a more convenient one). The suggestion that this work of Queneau’s be assigned to the Lullian combinatorial tradition is made not for the sake of claiming it as a “relative” of the algorithm but in order to set the latter in its proper place among Oulipian structures. By the way they basically function, these structures can be divided into two complementary categories. The first category is that of the proliferation of possibilities; the second is that of their reduction. In the first we find forms like Semo-Definitional Literature, the Lescurian word square, and attempts at “record setting” (Perec’s lipogram and palindrome). In the second are to be found, for instance, $S + 7$, phonetic translation, and antonymic poetry. In regard to our present subject, where *100,000 Billion Poems* uses a combinatorial system to extract a maximum of results from a limited amount of material, the algorithm—while still exploiting the potentiality of circular permutations—seeks to subject them to the harsh selectivity of a reductive form. Its aim is not to liberate potentiality but to coerce it.